

Department of Computer Science and  
Engineering,  
School of Technology,  
**Pandit Deendayal Energy University**  
**(PDEU)**  
**Software Engineering(24CS209P)**

Lab Experiment - 3

**Name:** Dax B Patel

**Roll no:** 24BCP269

# **Classes in Library Management System**

1. User
2. Student
3. Librarian
4. Notification
5. Transaction
6. Book
7. Report

## **List of methods and attributes for each class:**

### **1. User**

#### **Attributes:**

- userID : int
- name : String
- email : String
- phone : long int
- password : String

#### **Methods:**

- login()
- logout()
- viewProfile()

### **2. Student**

#### **Attributes:**

- department : String
- books\_borrowed : int

#### **Methods:**

- searchBook()

- borrowBook()
- returnBook()
- reserveBook()
- viewFines()
- viewHistory()

### 3. Librarian

**Attributes:**

- employeeId : int

**Methods:**

- addBook()
- updateBook()
- removeBook()
- issueBook()
- approveFine()
- acceptReturn()
- generateReport()

### 4. Notification

**Attributes:**

- notificationID : int
- notificationDate : Date
- message : String

**Methods:**

- sendNotification()

### 5. Transaction

**Attributes:**

- transactionID : int
- issueDate : Date
- dueDate : Date
- returnDate : Date
- fineAmount : float

**Methods:**

- calculateFine() : Float
- recordIssue()

- recordReturn()

## 6. Book

### Attributes:

- bookID : int
- title : String
- author : String
- category : String
- status : boolean
- publisher : String
- isbn : String

### Methods:

- checkAvailability()
- updateStatus()
- getDetails()

## 7. Report

### Attributes:

- reportID : int
- reportType : String
- generatedDate : Date

### Methods:

- generate()
- export()

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## Class Responsibility Cards (CRC):

### Class: User

#### Responsibilities:

- Authenticate users
- Maintain common user information

#### Collaborators:

- Student
- Librarian

## **Class: Student**

### **Responsibilities:**

- Search books
- Borrow, return, and reserve books
- View fines and transaction history

### **Collaborators:**

- Book
- Transaction
- Notification

## **Class: Librarian**

### **Responsibilities:**

- Manage book records
- Issue and accept return of books
- Approve fines
- Generate reports

### **Collaborators:**

- Book
- Transaction
- Report

## **Class: Book**

### **Responsibilities:**

- Store book details
- Track availability status

### **Collaborators:**

- Librarian
- Transaction

## **Class: Transaction**

### **Responsibilities:**

- Record book issue and return details
- Calculate fines
- Maintain borrowing history

**Collaborators:**

- Student
- Book

## **Class: Notification**

**Responsibilities:**

- Send notifications for due dates and fines

**Collaborators:**

- Student

## **Class: Report**

**Responsibilities:**

- Generate and export library reports

**Collaborators:**

- Librarian

# Class Diagram

